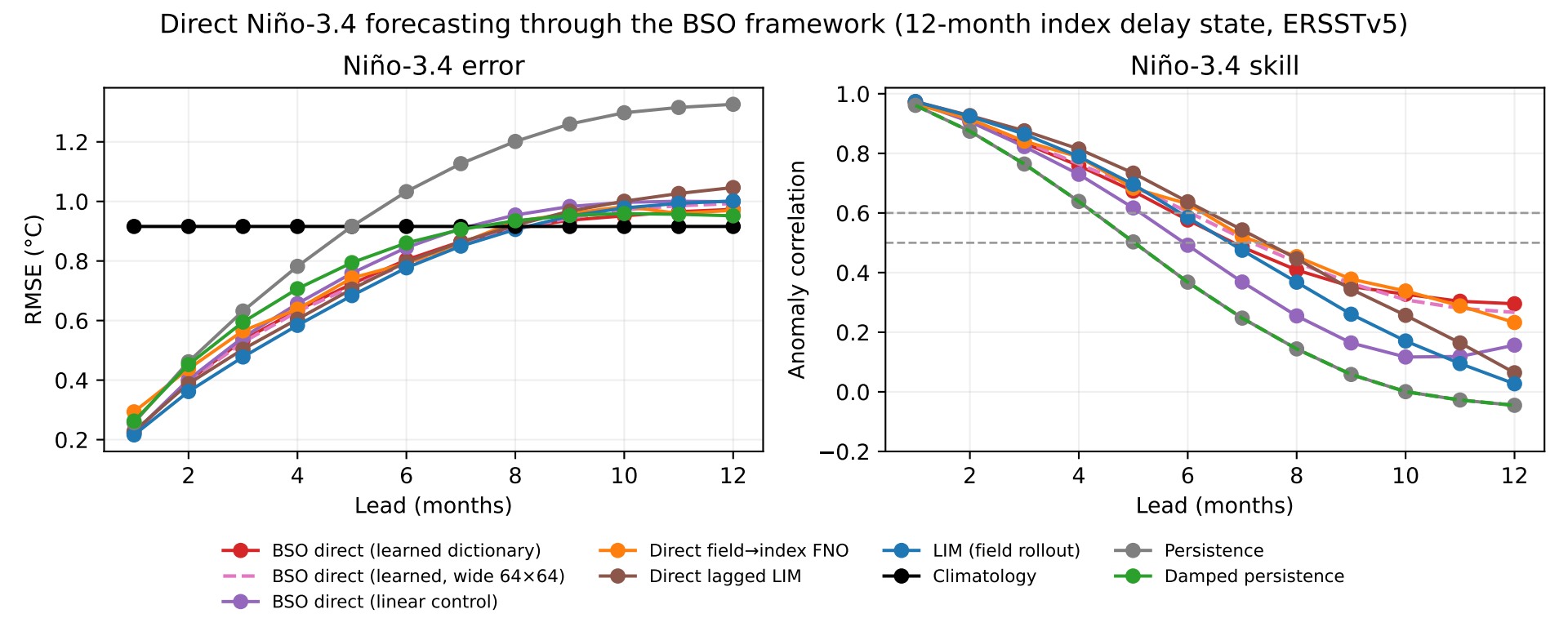
Direct Niño-3.4 forecasting through the BSO framework (12-month index delay state, ERSSTv5)
Niño-3.4 error
RMSE (°C)
Lead (months)
Niño-3.4 skill
Anomaly correlation
Lead (months)
BSO direct (learned dictionary)
BSO direct (learned, wide 64×64)
BSO direct (linear control)
Direct field→index FNO
Direct lagged LIM
LIM (field rollout)
Climatology
Persistence
Damped persistence